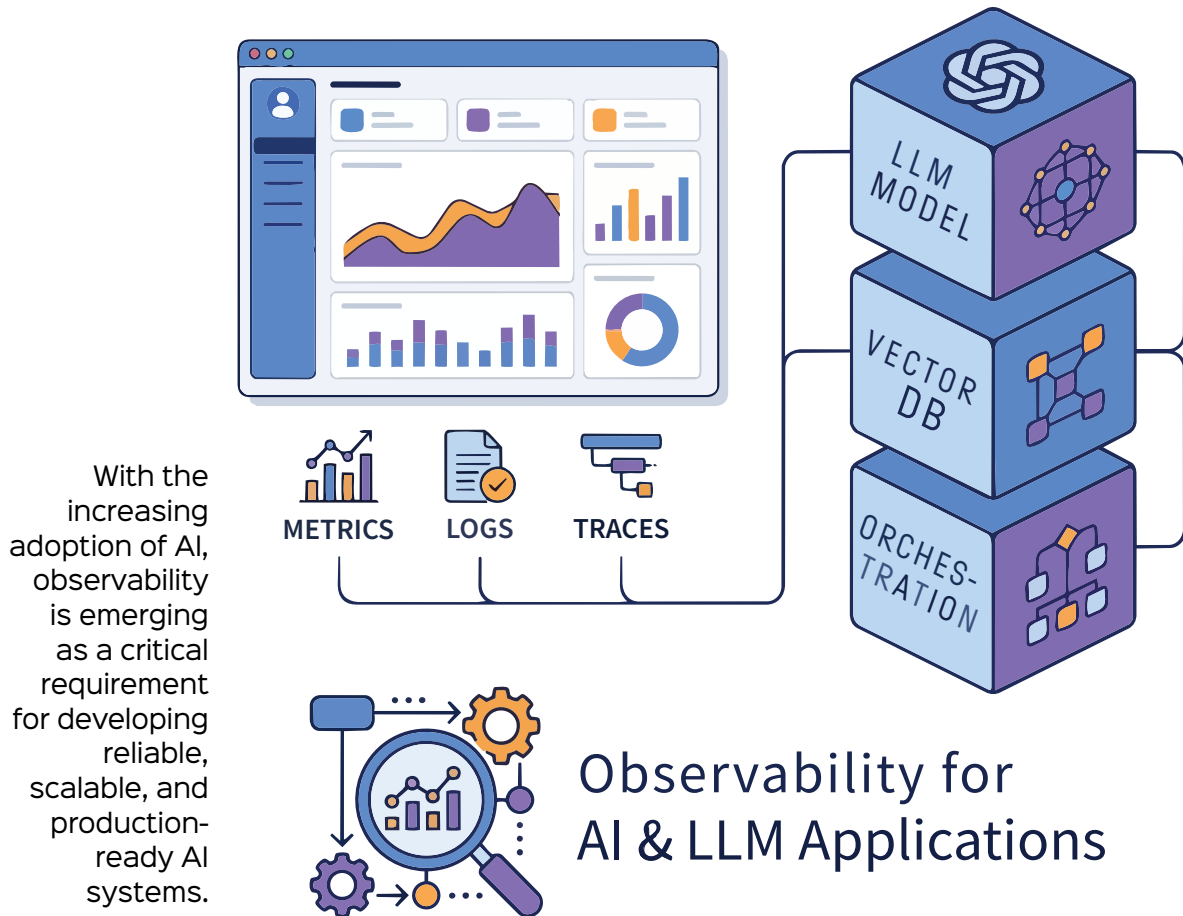


Why Observability is Vital for AI and LLM Applications



The rapid adoption of generative AI has transformed the creation and operation of modern applications. In contrast to conventional software systems, LLM-powered apps can reason over vast amounts of data, produce human-like responses, and communicate with users in natural language. Consequently, companies can now automate complicated tasks that previously required human intervention. However, a variety of operational issues start to surface when businesses apply these solutions in production settings.

In traditional applications, monitoring typically looks at system availability, error rates, application performance, and infrastructure health. To ensure reliable performance, teams monitor metrics such as CPU utilisation, memory consumption, request latency, uptime of services, and more.

These markers are important even though they only provide a limited picture of AI systems.

From an infrastructure perspective, an LLM application may be fully operational while simultaneously producing incorrect responses,

retrieving irrelevant data, producing inaccurate or misleading responses, or consuming excessive CPU power. In many instances, conventional monitoring approaches may demonstrate that the system is in excellent condition even when the end user experience is seriously affected.

This situation is unlike traditional software programs, which have deterministic outputs. LLMs have probabilistic outputs. The same question can have different answers depending on model parameters, retrieved context, prompt design, and